

CO5-AT2

Course Code: ITA0609

Course Title: Machine Learning

Student Name: S.Swagat

1).PSEUDOCODE ALGORITHM FOR TIME-SERIES DATA CLEANING AND TRANSFORMATION

Objective

To clean and transform time-series data by handling missing timestamps, interpolating missing values, removing outliers, resampling data at fixed intervals, and maintaining chronological consistency efficiently.

Pseudocode

ALGORITHM TimeSeriesDataCleaning

INPUT: Time-series dataset D

OUTPUT: Cleaned and transformed time-series dataset C

Step 1: Load the time-series dataset D.

Step 2: Convert the timestamp column into DateTime format.

Step 3: Remove records with invalid or missing timestamps.

Step 4: Set the timestamp column as the index of the dataset.

Step 5: Sort the dataset by the time index in ascending order.

Step 6: Remove duplicate timestamps if they exist.

Step 7: Check for missing timestamps between the minimum and maximum time.

Step 8: Create a complete time index with the required frequency.

Step 9: Reindex the dataset using the complete time index.

Step 10: Identify missing values created by missing timestamps.

Step 11: Interpolate the missing values using time-based interpolation.

Step 12: Detect outliers using a suitable method such as
IQR (Interquartile Range).

Step 13: Remove or replace the detected outlier values.

Step 14: Resample the cleaned data at a fixed interval
such as 1 hour, 30 minutes, or 1 day.

Step 15: Apply an aggregation function such as mean, sum,
minimum, or maximum during resampling.

Step 16: Sort the final dataset chronologically.

Step 17: Check for remaining missing values, duplicate timestamps,
and invalid values.

Step 18: Store or return the final cleaned dataset C.

END ALGORITHM

Explanation of Each Step

1. Indexing by Time

The timestamp column is converted into DateTime format and set as the index. This makes it easier to perform time-based operations such as interpolation and resampling.

2. Filtering Invalid Entries

Records with invalid or missing timestamps are removed. Duplicate timestamps are also identified and handled to maintain data consistency.

3. Handling Missing Timestamps

A complete time index is created between the minimum and maximum timestamps. The dataset is then reindexed so that missing timestamps become visible.

4. Interpolating Missing Values

Missing numerical values are filled using time-based interpolation. This estimates the missing value based on surrounding observations.

5. Removing Outliers

Outliers are detected using the IQR method. Values outside the lower and upper limits are considered outliers and can be removed or replaced.

The IQR limits are:

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR}$$

6. Resampling

The cleaned time-series data is resampled at a fixed interval such as hourly or daily.

For example:

Original: 10:01, 10:07, 10:14, 10:21

Resampled: 10:00, 10:15, 10:30, 10:45

An aggregation such as mean() can be used to calculate the value for each interval.

7. Maintaining Chronological Consistency

The final dataset is sorted by the timestamp index. This ensures that all records are arranged from the earliest time to the latest time.

8. Efficiency

For efficient processing:

- Convert timestamps only once.
- Use the timestamp as the DataFrame index.
- Sort the data only when required.
- Use vectorized Pandas operations instead of loops.
- Perform filtering before expensive operations.
- Use appropriate data types to reduce memory usage.
- Resample only after the data has been cleaned.

Conclusion

The proposed algorithm cleans time-series data by converting timestamps, removing invalid and duplicate entries, creating missing timestamps, interpolating missing values, detecting and removing outliers, and resampling the data at fixed intervals. Finally, the data is sorted chronologically and validated to ensure accuracy and consistency. Using Pandas vectorized operations makes the process efficient for large time-series datasets.

2).Feature Selection Algorithm Using Correlation-Based Filtering

Introduction

Feature selection is the process of selecting the most important features from a dataset while removing irrelevant and redundant attributes. It helps reduce dimensionality, improves model accuracy, decreases training time, and prevents overfitting. A correlation-based feature selection method evaluates the relationship between features and the target variable, removes highly correlated duplicate features, and retains only significant attributes.

Algorithm

Input:

- Dataset D containing features F1, F2, ..., Fn
- Target variable Y
- Correlation threshold CT (e.g., 0.80)
- Relevance threshold RT (e.g., 0.20)

Output:

- Optimized dataset with selected significant features

Detailed Pseudocode

BEGIN

Step 1: Load the dataset D.

Step 2: Preprocess the dataset.

 Remove missing values.

Remove duplicate records.

Step 3: Normalize all numerical features.

 FOR each feature F_i

$F_i = (F_i - \text{Min}(F_i)) / (\text{Max}(F_i) - \text{Min}(F_i))$

 END FOR

Step 4: Calculate feature relevance.

 FOR each feature F_i

$\text{Relevance}[F_i] = \text{Correlation}(F_i, \text{Target } Y)$

 END FOR

Step 5: Remove irrelevant features.

 FOR each feature F_i

 IF $|\text{Relevance}[F_i]| < RT$ THEN

 Remove F_i

 END IF

 END FOR

Step 6: Compute the feature-to-feature correlation matrix.

Step 7: Remove redundant features.

 FOR every pair (F_i, F_j)

 IF $|\text{Correlation}(F_i, F_j)| > CT$ THEN

 IF $|\text{Relevance}[F_i]| \geq |\text{Relevance}[F_j]|$ THEN

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        Remove Fj
    ELSE
        Remove Fi
    END IF
END IF
END FOR
```

Step 8: Create a new dataset D_{selected} with remaining features.

Step 9: Validate the selected features.

Split D_{selected} into Training and Testing sets.

Train a machine learning model.

Calculate Accuracy, Precision, Recall, and F1-score.

Compare results with the original dataset.

Step 10:

IF performance is improved or maintained THEN

Accept D_{selected}

ELSE

Adjust thresholds and repeat the process.

END IF

Step 11: Return D_{selected} .

END

Step-by-Step Explanation

Step 1: Load the Dataset

The algorithm begins by loading the dataset containing multiple input features and the target variable.

Step 2: Data Preprocessing

Before selecting features, the dataset is cleaned by:

- Removing missing values.
- Eliminating duplicate records.
- Ensuring data consistency.

This improves the quality of feature selection.

Step 3: Feature Normalization

All numerical features are normalized so that they have values between 0 and 1.

Normalization Formula:

$$X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

Where:

X_{norm} = Normalized value

X = Original feature value

X_{min} = Minimum value of the feature

X_{max} = Maximum value of the feature
Normalization prevents features with larger values from dominating the correlation calculations.

Step 4: Compute Feature Relevance

The correlation between each feature and the target variable is calculated using the Pearson Correlation Coefficient.

Pearson Correlation Formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \times \sum (y_i - \bar{y})^2]}}$$

Where:

r = Pearson correlation coefficient

x_i = Value of feature X

y_i = Value of target Y

$\bar{x}$ = Mean of X

$\bar{y}$ = Mean of Y

Σ = Summation of all values

Step 5: Remove Irrelevant Features

Features whose absolute correlation with the target is below the relevance threshold are removed.

Example:

- Threshold = 0.20
- Correlation = 0.05
- The feature is removed because it has very little influence on prediction.

Step 6: Create the Selected Dataset

After removing unnecessary features, a new dataset is created containing only significant attributes.

This reduced dataset has:

- Fewer features
- Less complexity
- Better computational efficiency

Step 7: Validate Model Performance

The selected features are validated by training a machine learning model.

Common evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-Score

If the reduced dataset performs equally well or better than the original dataset, the selected features are accepted.

Advantages of Correlation-Based Feature Selection

- Reduces dimensionality.
- Removes irrelevant features.
- Eliminates redundant attributes.
- Improves model accuracy.
- Reduces training time.
- Prevents overfitting.

- Makes models easier to interpret.
- Improves computational efficiency.

Applications

- Medical diagnosis
- Financial prediction
- Customer behavior analysis
- Image classification
- IoT and sensor data analysis
- Spam email detection

Conclusion

Feature selection using correlation-based filtering is an effective technique for reducing dataset dimensionality while preserving important information. By normalizing data, calculating feature relevance, removing low-impact features, eliminating highly correlated attributes, and validating model performance, the algorithm produces a simpler and more efficient dataset. This leads to faster training, reduced computational cost, improved prediction accuracy, and better overall machine learning performance.